



Getting Started with Open-Source Contributions: A Beginner's Guide

Jayapriya Pai, Red Hat
S Ashwin, Deutsche Bank

About Us



- ❑ Jayapriya Pai
- ❑ Senior Software Engineer, Red Hat
- ❑ Observability Engineering
- ❑ Open Source Contributor & Maintainer
- ❑ GSoC Mentor - Prometheus Operator
- ❑ Passionate Badmintonist
- ❑ <https://github.com/slashpai>



- ❑ S Ashwin
- ❑ Software Engineer, Deutsche Bank
- ❑ Open Source Contributor & Maintainer
- ❑ GSoC Mentee - Prometheus Operator
- ❑ Gym
- ❑ [LinkedIn](#)
- ❑ <https://github.com/AshwinSriram11>

Agenda

- What is Open Source and Open Source Community?
- Why contribute to Open Source?
- Pain points for Beginners
- How to Find the Right Project?
- Making your first contribution
- Mentorship programs
- Role of AI
- Networking and Community Groups
- Conclusion

Quick Show of Hands

- ❏ How many of you know what Open Source is?
- ❏ How many of you have done at least 1 contribution?
- ❏ How many of you are a regular contributor or want to become one?

Open Source

The Philosophy: The Cake & The Recipe



Proprietary Software

You get the cake. You can eat it, but you don't know what's in it.

recipe.js

```
class Cake {  
  ingredients = ['flour', 'sugar', 'eggs'];  
  bake() { ... }  
}
```

Open Source

You get the cake AND the recipe. You can modify the ingredients.



Study the Code



Use the Software



Change the Logic



Distribute to Others

Disclaimer: Image generated with Gemini

Open Source Community

“An open source community is a group of people (developers, users, and enthusiasts) who collectively build, maintain, and promote software with publicly accessible source code”

Community drives the project forward, shares knowledge and supports each other

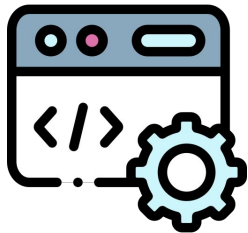
Eg: Kubernetes Community



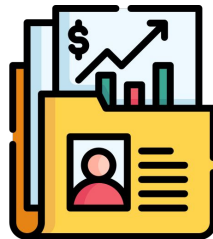
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Contributor Workflow

Why contribute to Open Source?



Learn by building
real-world software



Build a strong public
portfolio (proof of work)



Unlock career opportunities



Get feedback through
reviews & mentorship



Grow your network

Pain points for Beginners

- Finding beginner-friendly projects
- Setting up the project locally
- Understanding the codebase structure
- Picking an issue to work on
- Writing the code [*this is not a barrier now :)*]
- Opening a Pull Request
- Handling review comments

How to Find the Right Project?

1. Start with Interest

Pick something you actually enjoy / use.

2. Check Project Activity

Check recent commits and releases, PR reviews, issues getting replies.

3. Community Support

Active Slack/Discord channel, office hours, friendly discussions.

4. Good Docs and Beginner Friendly Issues -> Entrypoint

Good README.md, CONTRIBUTING.md, that helps you get to speed.

Issues Labels: Good First Issue, Help wanted

<https://landscape.cncf.io/>

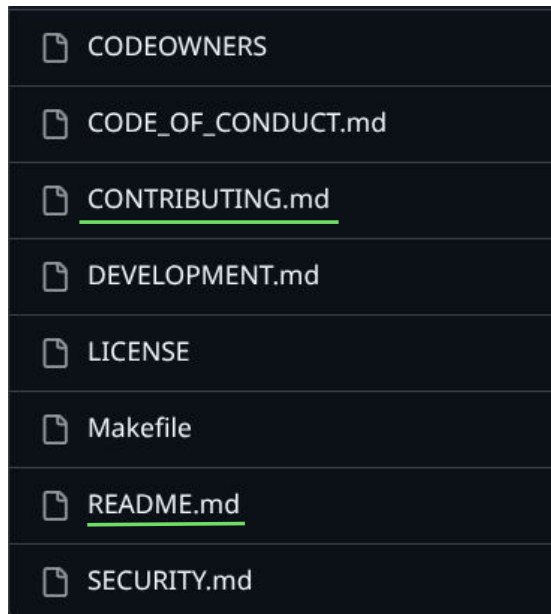
★ **Pro Tip - Start with CNCF Sandbox Projects**



Before Making your First Contribution



Step - 1



Step - 2



Step - 3



Code comes last. Context comes first.

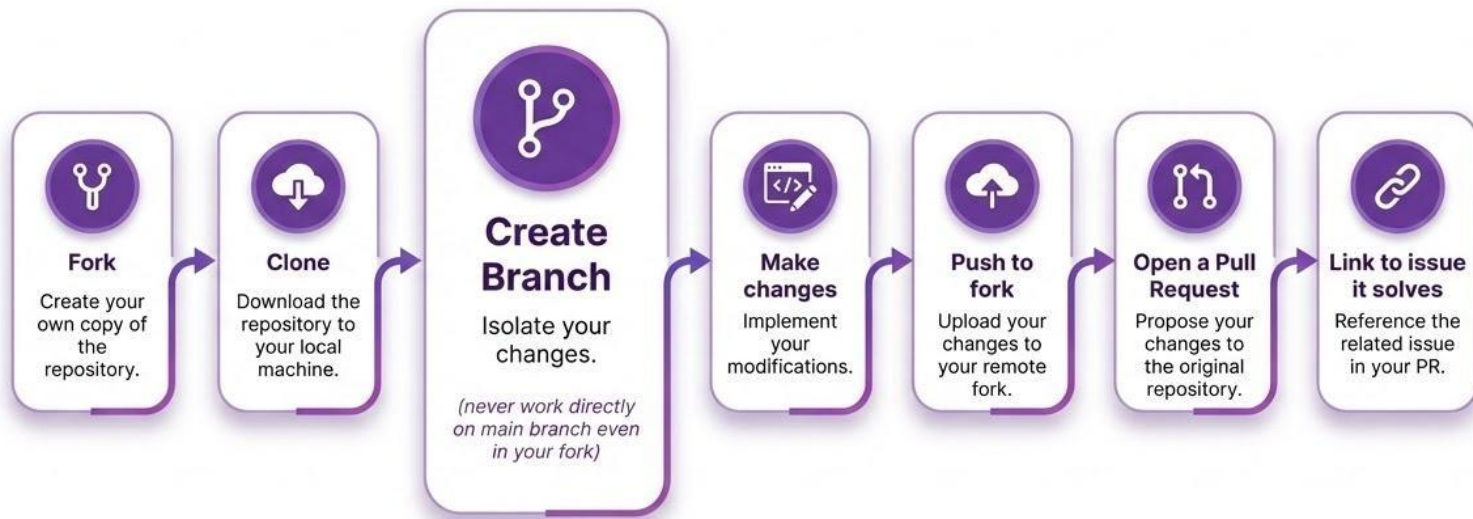
Making your first contribution

- Once you have selected a project follow these steps to contribute effectively:
- **Start Small:**
 - Do not aim for a massive feature refactor immediately
 - Small contributions like **fixing typos, improving documentation, or adding missing steps to a README** are valuable and low-risk
- **Communicate First:**
 - **Before writing code, comment on the issue** to let maintainers know you are interested to ensure your approach is correct and avoid duplicated work.

Contributor's Toolkit

- **Git and Github**
- **CI/CD Basics (Github Actions)**
- **Any language of your choice(Go, Rust,...)**
- **Passionate Mindset**
- **Clear communication**

Contribution Workflow



Disclaimer: Infographics image generated with Gemini

The background is a solid light purple color. Overlaid on this are several geometric shapes in different shades of purple. A large, dark purple circle is positioned on the left side. A medium-sized, medium-purple circle is in the center. A smaller, bright purple circle is on the right. A dark purple square is centered within the medium purple circle. The text 'Mentorship Programs' is written in white, bold, sans-serif font, centered horizontally and partially overlapping the medium purple circle and the dark purple square.

Mentorship Programs

Structured Mentorship Programs



- <https://summerofcode.withgoogle.com/>



- <https://lfx.linuxfoundation.org/tools/mentorship/>



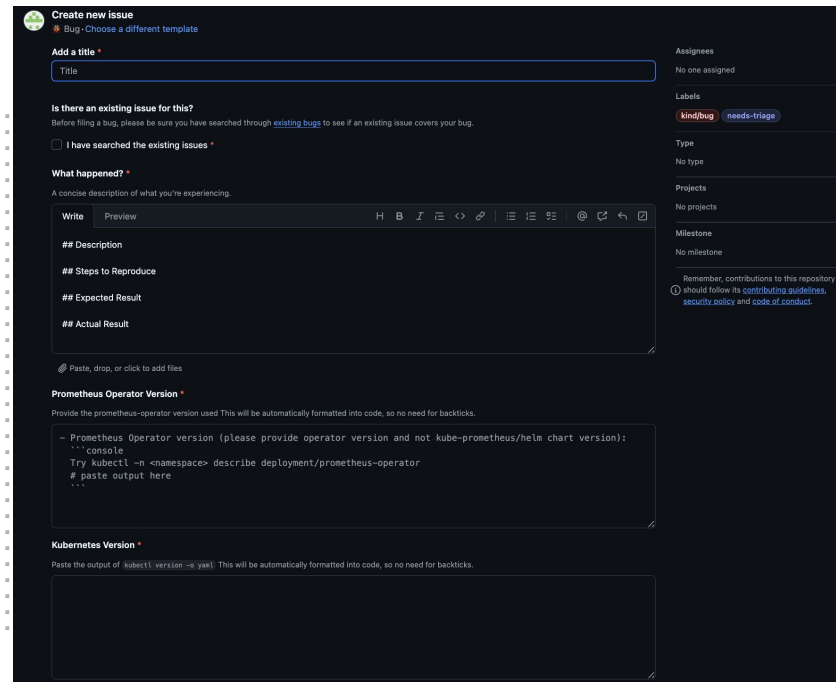
- <https://www.outreachy.org/>

Reviewer Workflow

Mastering the Issue

1. A Clear Title
2. Visual Proof
3. Steps to Reproduce
4. Environment Details

How to ask for Help?



The screenshot shows the 'Create new issue' page for the Prometheus Operator repository. The form includes a title field, a checkbox for 'I have searched the existing issues', and a 'What happened?' section with a rich text editor. The editor contains a description, steps to reproduce, expected result, and actual result. Below the editor, there are sections for 'Prometheus Operator Version' and 'Kubernetes Version', each with a code block for pasting the output of specific commands. The right sidebar shows 'Assignees' (No one assigned), 'Labels' (kind/bug, needs-triage), 'Type' (No type), 'Projects' (No projects), and 'Milestone' (No milestone). A note at the bottom of the sidebar reminds contributors to follow the repository's guidelines.

Create new issue

Bug - Choose a different template

Add a title *

Title

Is there an existing issue for this?

Before filing a bug, please be sure you have searched through [existing bugs](#) to see if an existing issue covers your bug.

☐ I have searched the existing issues *

What happened? *

A concise description of what you're experiencing.

Write Preview

H B I

Description

Steps to Reproduce

Expected Result

Actual Result

Paste, drop, or click to add files

Prometheus Operator Version *

Provide the prometheus-operator version used This will be automatically formatted into code, so no need for backticks.

```
- Prometheus Operator version (please provide operator version and not kube-prometheus/helm chart version):  
** console  
Try kubectl -n <namespace> describe deployment/prometheus-operator  
# paste output here  
...
```

Kubernetes Version *

Paste the output of kubectl version -o yaml. This will be automatically formatted into code, so no need for backticks.

Assignees

No one assigned

Labels

kind/bug needs-triage

Type

No type

Projects

No projects

Milestone

No milestone

Remember, contributions to this repository should follow its [contributing guidelines](#), [security policy](#) and [code of conduct](#).

The Art of the Pull Request (PR)

Keep it atomic: One Bug, One PR

A screenshot of a GitHub pull request interface. At the top, the title "The Bad PR" is displayed in a large, bold, black font. Below the title, the pull request is for a branch named "update-code" from the repository "bad-pr". The interface shows a diff view with a light blue header bar for the "update-code" branch. The diff shows changes to a file named "index.html". The changes are highlighted in a light blue box. The pull request is in a "Draft" state, indicated by a "Draft" label in the top right corner. The pull request has one commit, "update-code", and is associated with a pull request number "1". The pull request is open, as indicated by the "Open" button in the top right corner. The pull request has a "Description" section with the text "Fixed it." and a "Files Changed" section showing "45 files (Scope creep)". The "Status" section shows "No tests, no links". The pull request is created by a user named "bad-pr" and is assigned to a reviewer named "bad-pr". The pull request is open for review, as indicated by the "Review" button in the top right corner.

A screenshot of a GitHub pull request interface. At the top, the title 'The Stellar PR' is displayed in a large, bold, black font. Below the title, there's a header bar with a circular profile icon on the left, the text 'fix(ui): Add loader to auth button' in the center, and a green 'Draft' button on the right. Below the header bar, there's a light blue box containing the word 'Description' in bold. Underneath, the text 'Fixes #123 (Auto-close link)' is visible. Further down, the section '## Changes' is shown, followed by two list items: '[x] Added spinner component' and '[x] Updated tests'. At the bottom, there's a green box with the text 'Files Changed' in bold, a checkmark icon on the right, and '2 files (Atomic scope)' below it.

Disclaimer: Image generated with Gemini



Role of AI

How AI can help?

- **Understanding the codebase**
- **Asking more questions about the project if you didn't understand**
- **Generating Code and Documentation**
- **Removes the barrier to understand the project**
- **Creating notes (Eg: NotebookLM)**

What would I do if AI write code?

“Human in the loop”

Own the output Generative AI tools generated

- You have to still tell what to do and adjust the prompts till you get right solution
- Test Driven Development
 - First write/generate the tests to prove there needs to have a feature or a bug exists
 - Let the test fail
 - Have another commit as the solution

What you should not do with AI?

- Don't dump the AI generated code or documentation for the project maintainer to review
- ***Never push something which you don't understand :)***
 - Make sure you understand what the generated code does and adjust the code as required
- Use AI as a tool not as a replacement for human judgement

The background is a solid light purple color. Overlaid on this are several geometric shapes in different shades of purple. A large, dark purple circle is positioned on the left side. A medium-sized, medium-purple circle is on the right side. In the center, there is a square of a slightly darker purple shade. The text 'Networking & Community Groups' is centered horizontally and vertically, appearing in white.

Networking & Community Groups

Meetup Groups

<https://www.meetup.com/find/?categoryId=546&source=EVENTS>



meetup Search for anything Pune, IN English Log in Sign up

Events Groups

Technology Events near Pune, IN

Reset Any day Any type Within 30 kilometers

All events New Groups Social Activities Hobbies & Passions Sports & Fitness Travel & Outdoor Career & Business Technology Community & Environment Identity & Language Games Dancing Support & Coaching Music

DevConf.IN
open source community conference
Fri, Feb 13 - 9:00 AM IST
by AI Circle | ML Ops, LLMOps, AI Eng... • 4.8 ★
520 attendees

#02 AWS February Fusion - Pune
Sat, Feb 21 - 10:30 AM IST
by AWS User Group Pune • 4.7 ★
411 attendees

#03 : SheBuilds the Cloud: Rise of the Cloud Queens
Sun, Mar 8 - 10:30 AM IST
by AWS User Group Pune • 4.7 ★
157 attendees

AI SaaS Startup Meetup in Pune
Sun, Feb 15 - 11:00 AM IST
by eChai Pune Startup Network • 4.5 ★
142 attendees

AI Startup & Developers Meetup in Pune
Sat, Feb 14 - 4:00 PM IST
by eChai Pune Startup Network • 4.5 ★
195 attendees

Manufacturing Startup Meetup Pune
Sat, Feb 21 - 4:00 PM IST
by eChai Pune Startup Network • 4.5 ★
64 attendees

eChai Startup Demo Day in Pune
Sat, Feb 28 - 4:00 PM IST
by eChai Pune Startup Network • 4.5 ★
29 attendees

Dexclss Hackathon 2026
Sat, Feb 28 - 9:00 AM IST
by Dexclss Hackathon 2026
31 attendees

AI Business Automation Masterclass
Sat, Feb 21 - 12:00 PM IST
by Educity by NonceLabs - AI, Block... • 4.4 ★
370 attendees

Drupal Pune Meetup - February 21, 2026
Sat, Feb 21 - 9:30 AM IST
by Drupavers Association Pune • 4.7 ★
18 attendees

Beyond ELK: Elastic Stack in Practice
Sat, Mar 7 - 10:00 AM IST
by Elastic Pune User Group • 4.4 ★
8 attendees

Networking

CNCF Community: <https://community.cncf.io/>

Discover Community Groups

North America Latin America Europe Africa **Asia** Oceania Virtual CNCF Hosted Projects Community Days

Technical Community Groups

CLOUD NATIVE AHMEDABAD India	CLOUD NATIVE AURANGABAD, MH India	CLOUD NATIVE BAKU Azerbaijan	CLOUD NATIVE BANGALORE, KA India
CLOUD NATIVE BANGKOK Thailand	CLOUD NATIVE BANGLADESH Bangladesh	CLOUD NATIVE BHUBANESWAR, OD India	CLOUD NATIVE CALICUT, KL India
CLOUD NATIVE CHANDIGARH, CH India	CLOUD NATIVE CHENNAI, TN India	CLOUD NATIVE COIMBATORE, TN India	CLOUD NATIVE COMMUNITY JAPAN, TOKYO Japan
CLOUD NATIVE DEHRADUN, UT India	CLOUD NATIVE DURGAPUR, WB India	CLOUD NATIVE GANDHINAGAR, GJ India	CLOUD NATIVE GURUGRAM, HR India
CLOUD NATIVE HANOI Vietnam	CLOUD NATIVE HO CHI MINH, HO CHI MINH CITY Vietnam	CLOUD NATIVE HOOGHLY, WB India	CLOUD NATIVE HYDERABAD India
CLOUD NATIVE ISLAMABAD Pakistan	CLOUD NATIVE JAIPUR, RJ India	CLOUD NATIVE JALGAON, MH India	CLOUD NATIVE JORDAN, AMMAN GOVERNORATE Jordan
CLOUD NATIVE KANPUR, UTTAR PRADESH India	CLOUD NATIVE KATHMANDU Nepal	CLOUD NATIVE KOCHI, KL India	CLOUD NATIVE KOLKATA, WB India
CLOUD NATIVE KOTTAYAM, KL India	CLOUD NATIVE KUALA LUMPUR, FEDERAL TERRITORY OF KUALA LUMPUR Malaysia	CLOUD NATIVE LAHORE, PUNJAB Pakistan	CLOUD NATIVE LUCKNOW, UP India
CLOUD NATIVE MANILA, NCR Philippines	CLOUD NATIVE MUMBAI CITY, MH India	CLOUD NATIVE NAGPUR, MH India	CLOUD NATIVE NASHIK India
CLOUD NATIVE NEW DELHI, DL India	CLOUD NATIVE NOIDA, UP India	CLOUD NATIVE PUNE India	CLOUD NATIVE RAJKOT, GJ India

Discover Upcoming events

Search for a city to find events

Filter by: Application Frameworks Cloud Native AI Data Architecture Embedded Functions General Orchestration Inclusion Languages

February Virtual Meetup - Transforming KServe Into a Zero Trust

Hyderabad Feb 12

Kubernetes Bangalore Meetup

Bengaluru Feb 28

Managing AI/ML Model Provenance and Compliance with KitOps

Kochi Feb 18

KCD Kochi 2026

Kochi Apr 11

Build your own local community



Conclusion

- What is Open Source and why contribution matters?
- Best practices
- Contributor and Reviewer workflows

Your Mission Today:

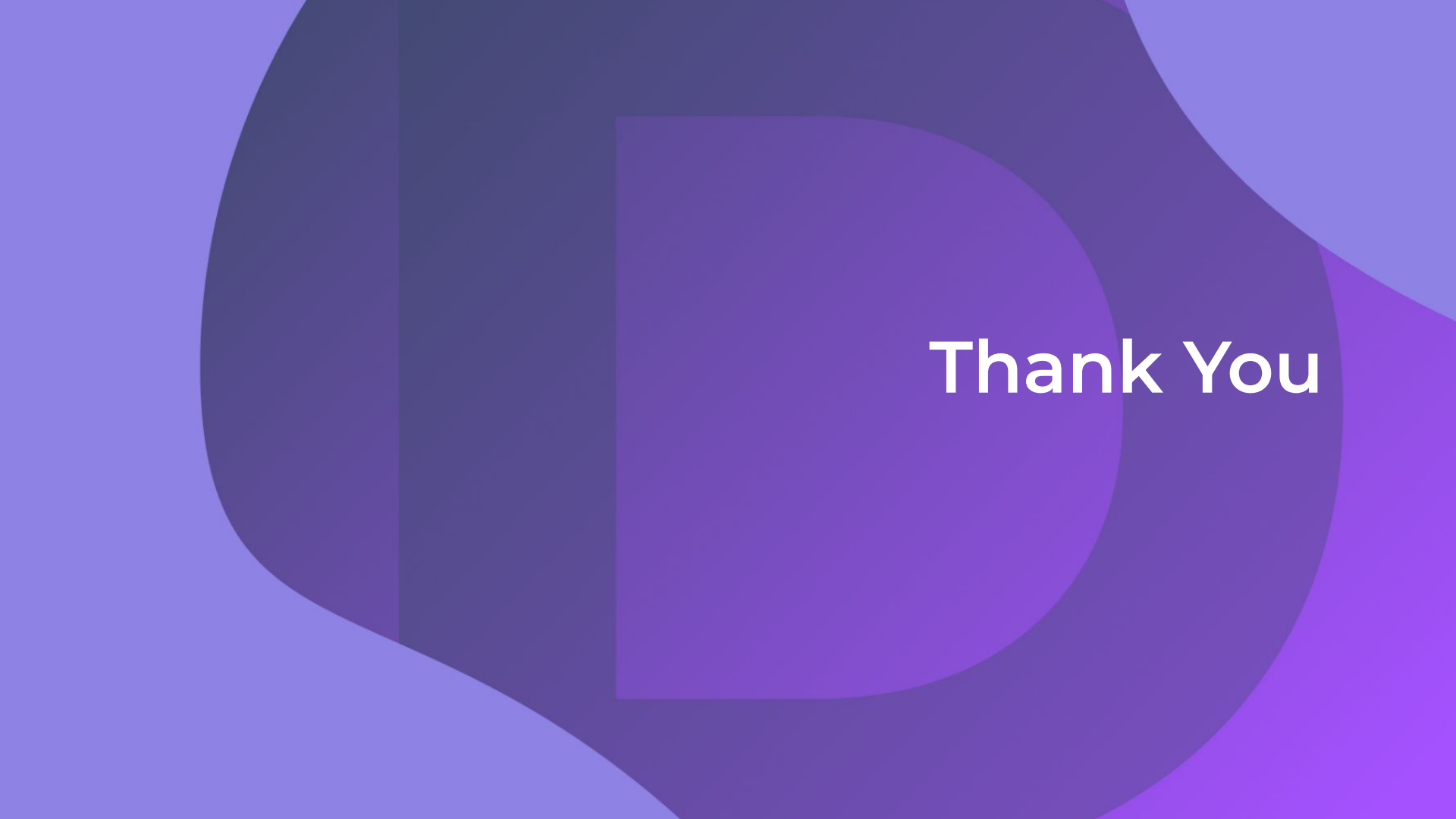
1. Find a repo you use.
2. Find a typo or broken link.
3. Fix it.

Welcome to the team.



Helpful Resources

- <https://community.cncf.io/>
- <https://clotributor.dev/search>
- <https://github.com/MunGell/awesome-for-beginners>
- https://www.youtube.com/watch?v=mklEhT_RLos
- <https://landscape.cncf.io/>
- <https://github.com/open-source>
- <https://github.blog/open-source>
- <https://github.com/topics/hacktoberfest>
- <https://www.linuxfoundation.org/resources/open-source-guides/participating-in-open-source-communities>

The background is a solid light purple color. Overlaid on this are several large, overlapping circles in various shades of purple, ranging from a very light lavender to a deep, dark indigo. The circles are arranged in a way that they partially obscure each other, creating a layered effect. In the center-right area, the words "Thank You" are written in a clean, white, sans-serif font. The text is positioned over a medium-dark purple circle, which makes it stand out clearly.

Thank You